Shahzaib Saqib Warraich

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AI researcher working on interpretability and evaluation of language models: understanding what models actually know and building rigorous methods to measure it. Publications at ICLR 2026 (Oral) and COLM 2026; Fulbright Scholar; co-founder of an applied AI firm, with prior industry experience shipping production ML systems.

Education

University of Southern California Los Angeles, CA
M.S. in Applied Data Science, Viterbi School of Engineering Jan 2024 – Dec 2025
Fulbright Scholar (U.S. Department of State, 2023 cohort); **USC Andrew J. Viterbi Scholarship** (2024–2025).
Coursework: Applied NLP, Applied Machine Learning, Deep Learning, Analysis of Algorithms, Data Management.

National University of Sciences and Technology (NUST) Islamabad, Pakistan
B.E. in Electrical Engineering and Computer Science Sep 2017 – Jun 2021

Publications

[1] **How Reliable is Language Model Micro-Benchmarking?**
G. Yauney, **S. S. Warraich**, S. Swayamdipta. *ICLR 2026* **(Oral)**. [paper] [code]

[2] **Sample, Align, Synthesize: Graph-Based Response Synthesis with ConGrs**
S. Ghosh, **S. S. Warraich**, D. Tarsadiya, G. Yauney, S. Swayamdipta. *COLM 2026*; *ER@NeurIPS 2025*. [paper] [code]

[3] **Residual Balanced Attention Network for Real-Time Traffic Scene Semantic Segmentation**
A. Kherraki, **S. S. Warraich**, M. Maqbool, R. El Ouazzani. *IJECE* 2023. [paper]

Research Experience

USC NLP Group, DILL Lab (Data, Interpretability, Language and Learning) Los Angeles, CA
Graduate Research Assistant, Thomas Lord Department of Computer Science Jan 2024 – Jan 2026
Advisor: Prof. Swabha Swayamdipta. Mentors: Dr. Gregory Yauney, Matthew Finlayson, Sayan Ghosh.

Micro-benchmark reliability ICLR 2026 Oral

- Proposed and implemented confidence-based stratified sampling for micro-benchmark selection, adapting a sampling technique from the computer vision literature to language model evaluation.
- Built infrastructure to compare micro-benchmark selection methods: stratified sampling, tinyBenchmarks (P-IRT, GP-IRT), and task-reduction baselines.
- Maintained and extended the project codebase for reproducible meta-evaluation experiments.

Consensus Graphs (ConGrs) for response synthesis COLM 2026

- Applied the consensus-decoding framework to factuality-critical tasks (long-form biographies, PopQA) with five replications per model, establishing baselines and characterizing the factuality–informativeness trade-off.
- Built high-throughput batch-processing infrastructure for FActScore evaluation; added response caching that removed the main processing bottleneck.
- Designed ablations on how sample size affects the alignment and synthesis stages, plus a sampling-temperature sensitivity analysis, to characterize the framework's robustness.
- Extended the framework to refusal-aware settings across three domains (numerical false presuppositions, scientific attributions, historical events), using LLM-as-a-judge abstention detection.

Industry Experience

Turon AI Hybrid
Co-Founder & COO (Strategic Partnerships, Customer Success) Jan 2026 – Present
Co-Founder & CTO (Technical Leadership, Solution Architecture) Jun 2024 – Dec 2025
Applied AI R&D and production engineering for manufacturing, automotive, and construction clients.

- Own solution architecture and technical pre-sales: lead client discovery, scope system designs, and

convert them into signed engagements and delivery plans.
- Architect and ship production agentic systems and AI infrastructure for enterprise clients.
- Built and run the firm's commercial and operating layer: entity structure, go-to-market, engineering infrastructure, and technical hiring.

Paramount Los Angeles, CA
Data Scientist Intern May 2025 – Oct 2025

- Improved content-performance forecasting accuracy by up to **20%** across key business KPIs by integrating multi-platform social-media engagement signals into the predictive models.
- Built a production GenAI explainability system using an ensemble of LLM judges with majority voting to attribute performance at the title level, making model insights usable by non-technical stakeholders.

Retrocausal Redmond, WA (Remote)
AI Engineer II Dec 2022 – Dec 2023

- Designed and deployed assembly-verification models (object detection, activity recognition) reaching **98%** detection accuracy for Fortune 500 clients incl. Schneider Electric, Carrier, and Honda (Assembly Copilot).
- Built an automated model-training pipeline using cut-paste-mix augmentation to generate synthetic training data, cutting manual labeling by **80%** while maintaining model performance.
- Conceived and led development of a computer-vision ergonomics risk-assessment module, a new product capability for workplace safety (Kaizen Copilot).

Adlytic AI Lahore, Pakistan
AI Engineer I Jun 2021 – Dec 2022

- Reduced edge inference latency by **66%** with knowledge distillation and NVIDIA TensorRT optimization, staying within **1.5%** of baseline accuracy in production.
- Built and deployed a cloud-based recommender system with **97%** prediction accuracy, contributing to a **3**× increase in product conversion rates.

Teaching

University of Southern California, Viterbi School of Engineering Los Angeles, CA
Course Producer, CSCI 544: Applied Natural Language Processing (graduate) Aug 2024 – May 2025

- Produced course content and supported instruction: Fall 2024 (Prof. Swabha Swayamdipta), Spring 2025 (Dr. Xuezhe Ma).

Leadership

Viterbi Graduate Student Association (VGSA), USC Los Angeles, CA
Vice President of Outreach May 2025 – Dec 2025
Senator Aug 2024 – Apr 2025

- Represented nearly **6,000** Viterbi graduate students as primary liaison to university administration.
- Managed internal and external partnerships to expand graduate student resources and professional opportunities.
- Delivered monthly reports to the Viterbi School Dean on student concerns, with actionable recommendations.

Writing

Dawn Remote
Op-Ed Contributor, Technology & Infrastructure Aug 2026 – Present
Op-eds for Pakistan's most-read English daily: *Pakistan's AI future*; *The data centre and water question* [author page]

Skills

Programming: Python, C/C++, SQL
ML / LLMs: PyTorch, Hugging Face, vLLM, scikit-learn, OpenCV, NVIDIA TensorRT
Data: Pandas, Dask, PySpark, Matplotlib, Seaborn, Plotly
Cloud & Infra: AWS SageMaker, GCP Vertex AI, Docker, Kubernetes, MongoDB, Git
Languages: English (professional), Urdu (native), French (DELF A2)